

Mingjian(Norman) Li

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Education

University of Pennsylvania	Aug 2026 (Incoming)
• MSE in Computer and Information Science	
New York University	May 2026
• B.S. in Computer Science; Minor in Mathematics	GPA: 3.97/4.0

Experience

Robotics Software Engineer Intern, Zhiyuan Company – Chengdu, China	Dec 2025 – Feb 2026
• Built vehicle detection and high-precision localization module for automated parking system, fused RGB camera data with Livox Mid-360 LiDAR point cloud to estimate 3D vehicle position and orientation for parking guidance.	
• Developed modular ROS-based software with YOLO real-time detection, optimized sensor sync and coordinate transformation, achieving $\pm 5\text{cm}$ localization accuracy in dynamic parking environments.	
Software Engineer Intern, SeeM(useums)– Pittsburgh	Dec 2024 – May 2025
• Developed a full-stack application using React, Next.js, Redux, and Flask, implementing authentication, project management, and AI agent integration, deployed on AWS for scalable infrastructure.	
• Built a 3D reconstruction pipeline based on 3D Gaussian splatting and SuGaR, supporting multiple data formats and enabling interactive visualization with Three.js.	

Publication

[1] **Mingjian Li***, Shihang Wei*, Ran Gong, Yueyu Hu, Yao Wang. “DanceNet3D: A 3D Dance Dataset with Multi-View Videos and 3DGS Reconstructions.” *CVPR 2026 3DMV Workshop*. Project Page

Research

Deep Learning based Accelerated MR Image Reconstruction NYU (Mentor: Haoyang Pei)	Spe 2025 – Dec 2025
• Conducting research on deep learning–based MR image reconstruction to accelerate scan times, evaluating multiple (CNN, UNet, Variational Networks, Diffusion Models) on the FastMRI dataset for high-quality reconstruction from undersampled k-space data.	
• Refined PyTorch and CUDA implementations for multi-GPU HPC , enabling distributed training with synchronized gradient updates, resulting in a $4\times$ speedup in performance and reconstruction fidelity.	
4D Human Dance Reconstruction NYU Video Lab (Advisor: Prof. Yao Wang)	May 2025 – Present
• Developing a multiview, multiframe human dancer dataset and a full pipeline for data processing, 3D Gaussian construction, mesh reconstruction, and parameter tuning, while replicating and benchmarking advanced 3DGS methods to achieve high-quality, geometrically consistent pose reconstructions (PSNR > 35).	
• Developed a SAM3 -powered video and image compression pipeline achieving up to 230$\times$ reduction, and implemented real-time 3D rendering modules in Unity and Unreal Engine .	

Projects

SO-ARM101 Robot Learning Pipeline	Spring 2026
• 3D printed and assembled an SO-ARM101 robotic arm, calibrated Feetech servo motors, and configured leader-follower teleoperation for robot control and demonstration collection.	
• Building a robot learning pipeline by collecting teleoperation data for policy training and developing simulation environments in Isaac Lab for sim-to-real experimentation.	
ML Systems Implementation	Fall 2025
• GPU Tensor Operators — Implemented core tensor ops (elementwise, matmul, reduction, cross-entropy) in CUDA/C++, optimizing kernels on NVIDIA T4 (CUDA 12.8) with full unit-test correctness.	
• MLP with Auto-Diff — Built a Python-based deep learning system via pybind11 ; implemented AGTensor for dynamic graphs and backprop, training a 2-layer MLP on MNIST (97% accuracy).	